

Limit points of the finite-volume ground states of $P(\varphi)_2$ are ground states of the quasi-local dynamics

Milestone M0

Abstract

Let $H(g)$ be the $P(\varphi)_2$ Hamiltonian with spatial cutoff g , let Ω_g be its (unique) ground state, and let $\omega_g = \langle \Omega_g, \cdot \Omega_g \rangle$ be the induced state on the time-zero quasi-local algebra $\mathfrak{A}$. Let α be the Glimm–Jaffe automorphism group of $\mathfrak{A}$, which is independent of g . We prove that every limit point of the net $(\omega_g)_{g \uparrow 1}$, in the topology of norm convergence on local algebras, is a *ground state* of the C^* -dynamical system $(\mathfrak{A}, \alpha)$.

The proof is an exact identity, not an estimate: it produces no error terms and requires no hypotheses beyond facts already established by Glimm and Jaffe. Consequently the removal of the spatial cutoff is reduced, modulo the compactness already available from [4, Thm. 2.3], to the *uniqueness* of the ground state of $(\mathfrak{A}, \alpha)$.

We also record two invariance properties obtained at no cost: the set of limit points is stable under space translations (so uniqueness forces translation invariance, with no space-averaging), and, for even P , every ω_g is $\mathbb{Z}_2$ -invariant.

1 Setting

1.1 Free field and local algebras

Let $\mathcal{H}_0 = \Gamma_s(L^2(\mathbb{R}))$ be the symmetric Fock space over $L^2(\mathbb{R})$ and $\mu = (-\partial_x^2 + m_0^2)^{1/2}$ with $m_0 > 0$. For real $f \in H^{-1/2}(\mathbb{R})$ and real $h \in H^{1/2}(\mathbb{R})$ the time-zero fields $\varphi_0(f)$ and $\pi_0(h)$ are essentially self-adjoint on the finite-particle domain [6, Prop. 1.2.6]; we denote their closures by the same symbols and set

$$W(f, h) := \exp(i(\varphi_0(f) + \pi_0(h))),$$

a unitary on $\mathcal{H}_0$. For bounded open $O \subseteq \mathbb{R}$ put

$$\mathfrak{A}(O) := \{W(f, h) : f, h \in C_c^\infty(O) \text{ real}\}'' \subseteq B(\mathcal{H}_0), \quad \mathfrak{A} := \overline{\bigcup_O \mathfrak{A}(O)}^{\|\cdot\|}.$$

The net $O \mapsto \mathfrak{A}(O)$ is isotonomous, so $\mathfrak{A}$ is a C^* -algebra. For $r > 0$ we write $O_r := \{x \in \mathbb{R} : \text{dist}(x, O) < r\}$.

Let τ_a denote the automorphism of $\mathfrak{A}$ induced by the space translation $U_a = \Gamma(e^{ia \cdot})$, so $\tau_a(\mathfrak{A}(O)) = \mathfrak{A}(O + a)$.

1.2 The cutoff Hamiltonian

Let P be a real polynomial, bounded below, of even degree, and let

$$\mathcal{G} := \{g \in C_c^\infty(\mathbb{R}) : 0 \leq g \leq 1, g \equiv 1 \text{ on some nonempty bounded open interval}\}.$$

For $g \in \mathcal{G}$ set

$$H(g) := H_0 + \int_{\mathbb{R}} :P(\varphi_0(x)) : g(x) dx - E(g), \quad E(g) := \inf \text{spec} \left(H_0 + \int :P(\varphi_0) : g \right).$$

Fact 1.1 (Glimm–Jaffe; Rosen). $H(g)$ is self-adjoint and bounded below, and $H(g) \geq 0$ with $\inf \text{spec } H(g) = 0$. [2, 8, 5]

Fact 1.2 (Glimm–Jaffe). 0 is a *simple and isolated* eigenvalue of $H(g)$. Its normalised eigenvector Ω_g may be chosen strictly positive in the Q -space (Schrödinger) representation. [3, §2.2–2.3]

We write $\omega_g := \langle \Omega_g, \cdot \Omega_g \rangle$, a state on $\mathfrak{A}$.

Fact 1.3 (Glimm–Jaffe: locality and g -independence). There is a one-parameter group $(\alpha_t)_{t \in \mathbb{R}}$ of $*$ -automorphisms of $\mathfrak{A}$ such that

- (i) $\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ for every bounded open O and every $t \in \mathbb{R}$;
- (ii) if $A \in \mathfrak{A}(O)$ and $g \in \mathcal{G}$ satisfies $g \equiv 1$ on $O_{|t|}$, then $\alpha_t(A) = e^{itH(g)} A e^{-itH(g)}$.

In particular α does not depend on g . [2, §III], [3, §3.5–3.6], [6, §4.1]

Property (i) is an *exact* light cone: there is no tail, in contrast with the Lieb–Robinson bounds available for lattice systems. This is used decisively in §4.

1.3 The directed set

Order $\mathcal{G}$ by

$$g \preccurlyeq g' \quad :\Longleftrightarrow \quad \{g = 1\}^\circ \subseteq \{g' = 1\}^\circ,$$

and for bounded open O put

$$d(g, O) := \text{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ) \in (0, \infty].$$

Lemma 1.4. $(\mathcal{G}, \preccurlyeq)$ is directed; it is stable under translations, and for each $a \in \mathbb{R}$ the map $g \mapsto \tau_a g$ is an order isomorphism of $\mathcal{G}$ with cofinal range. Moreover, for every bounded open O and every $R > 0$ there is $g_0 \in \mathcal{G}$ with $d(g, O) > R$ for all $g \succcurlyeq g_0$.

Proof. Given $g_1, g_2 \in \mathcal{G}$, the sets $\{g_i = 1\}^\circ$ are bounded and open; choose a bounded open interval I containing their closures and $g_3 \in C_c^\infty(\mathbb{R})$ with $0 \leq g_3 \leq 1$ and $g_3 \equiv 1$ on I . Then $g_3 \in \mathcal{G}$ and $g_3 \succcurlyeq g_1, g_2$. Translation stability and the order-isomorphism property are immediate from $\{\tau_a g = 1\}^\circ = \{g = 1\}^\circ + a$; cofinality follows from directedness. For the last claim choose $g_0 \equiv 1$ on a bounded open interval containing $\overline{O_R}$; then $g \succcurlyeq g_0$ gives $\{g = 1\}^\circ \supseteq O_R$, hence $d(g, O) \geq R$; enlarging g_0 slightly gives the strict inequality. $\square$

2 Ground states of $(\mathfrak{A}, \alpha)$

Because $\mathfrak{A}$ contains Weyl unitaries, $t \mapsto \alpha_t(A)$ is in general *not* norm-continuous: two Weyl unitaries $W(f, h), W(f', h')$ with $(f, h) \neq (f', h')$ are at distance 2 in the CCR C^* -algebra. Hence α has no densely defined norm-generator and the familiar characterisation “ $-i\omega(A^*\delta(A)) \geq 0$ ” is unavailable. We therefore use the spectral form throughout. (See Remark 5.1.)

Definition 2.1. A state ω on $\mathfrak{A}$ is a *ground state* of $(\mathfrak{A}, \alpha)$ if

- (a) $\omega \circ \alpha_t = \omega$ for all $t \in \mathbb{R}$; and

- (b) $t \mapsto \omega(A \alpha_t(B))$ is continuous for all $A, B \in \mathfrak{A}$, and for every $f \in L^1(\mathbb{R})$ whose inverse Fourier transform

$$\check{f}(E) := \int_{\mathbb{R}} f(t) e^{itE} dt$$

vanishes on $[0, \infty)$, one has $\int_{\mathbb{R}} f(t) \omega(A \alpha_t(B)) dt = 0$.

The continuity requirement in (b) is what makes the integral meaningful; it is verified independently in Theorem 4.2, so it is not an extra assumption there.

Lemma 2.2. *Let ω satisfy (a). Then (b) holds if and only if, in the GNS triple $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$, the unitary group U implementing α with $U(t)\Omega_\omega = \Omega_\omega$ has self-adjoint generator $H_\omega \geq 0$. In particular Definition 2.1 agrees with the usual notion [1, Def. 5.3.18].*

Proof. By (a) the group $U(t)\pi_\omega(A)\Omega_\omega := \pi_\omega(\alpha_t(A))\Omega_\omega$ is well defined, unitary and weakly, hence strongly, continuous; write $U(t) = e^{itH_\omega}$. For $A, B \in \mathfrak{A}$ set $u = \pi_\omega(A)^*\Omega_\omega$, $v = \pi_\omega(B)\Omega_\omega$ and let $\rho := \langle u, P_{H_\omega}(\cdot)v \rangle$, a complex measure on $\mathbb{R}$ of finite total variation $|\rho|(\mathbb{R}) \leq \|u\|\|v\|$. Using $U(t)^*\Omega_\omega = \Omega_\omega$,

$$\omega(A \alpha_t(B)) = \langle u, U(t)v \rangle = \int_{\mathbb{R}} e^{itE} d\rho(E).$$

Fubini (legitimate since $f \in L^1$ and $|\rho|(\mathbb{R}) < \infty$) gives $\int f(t) \omega(A \alpha_t(B)) dt = \int_{\mathbb{R}} \check{f}(E) d\rho(E)$. If $H_\omega \geq 0$ then ρ is supported in $[0, \infty)$, where $\check{f}$ vanishes, so the integral is 0. Conversely, if the integral vanishes for all such f , then, since $\{\check{f} : f \in L^1, \check{f}|_{[0, \infty)} = 0\} \supseteq C_c^\infty((-\infty, 0))$, we get $\rho|_{(-\infty, 0)} = 0$ for all A, B ; as vectors of the form $\pi_\omega(B)\Omega_\omega$ are dense, $P_{H_\omega}((-\infty, 0)) = 0$. $\square$

3 Two invariances, for free

Lemma 3.1 (local α -invariance). *Let O be bounded open, $A \in \mathfrak{A}(O)$, $g \in \mathcal{G}$ and $|t| < d(g, O)$. Then $\omega_g(\alpha_t(A)) = \omega_g(A)$.*

Proof. Since $|t| < d(g, O)$ we have $O_{|t|} \subseteq \{g = 1\}^\circ$, so $g \equiv 1$ on $O_{|t|}$ and Fact 1.3(ii) gives $\alpha_t(A) = e^{itH(g)} A e^{-itH(g)}$. As $H(g)\Omega_g = 0$ we have $e^{-itH(g)}\Omega_g = \Omega_g$, whence $\omega_g(\alpha_t(A)) = \langle \Omega_g, e^{itH(g)} A e^{-itH(g)} \Omega_g \rangle = \langle \Omega_g, A \Omega_g \rangle$. $\square$

Lemma 3.2 (translation covariance of the net). *For every $a \in \mathbb{R}$ and $g \in \mathcal{G}$ one has $\omega_{\tau_a g} = \omega_g \circ \tau_{-a}$.*

Proof. U_a implements τ_a and commutes with H_0 , while $U_a(\int : P(\varphi_0) : g) U_a^* = \int : P(\varphi_0) : \tau_a g$. Hence $H(\tau_a g) = U_a H(g) U_a^*$ and $E(\tau_a g) = E(g)$. By Fact 1.2 the ground state of $H(\tau_a g)$ is $U_a \Omega_g$ up to a phase, which does not affect the induced state. Therefore $\omega_{\tau_a g}(A) = \langle U_a \Omega_g, A U_a \Omega_g \rangle = \langle \Omega_g, U_a^* A U_a \Omega_g \rangle = \omega_g(\tau_{-a}(A))$. $\square$

Lemma 3.3 ($\mathbb{Z}_2$ invariance). *Assume P is even and let θ be the automorphism of $\mathfrak{A}$ with $\theta(W(f, h)) = W(-f, -h)$. Then $\omega_g \circ \theta = \omega_g$ for every $g \in \mathcal{G}$.*

Proof. θ is implemented by $\Theta = \Gamma(-1)$, which is unitary, self-adjoint, commutes with H_0 , and satisfies $\Theta : P(\varphi_0(x)) : \Theta^* = : P(\varphi_0(x)) :$ because P is even. Hence $\Theta H(g) \Theta^* = H(g)$, so $\Theta \Omega_g$ is a normalised ground state and, by simplicity, $\Theta \Omega_g = c \Omega_g$ with $c \in \{\pm 1\}$ (using $\Theta^2 = 1$). In the Q -space representation Θ acts by $(\Theta \psi)(q) = \psi(-q)$, so $\Theta \Omega_g > 0$ pointwise by Fact 1.2; hence $c = +1$ and $\omega_g(\theta(A)) = \langle \Theta \Omega_g, A \Theta \Omega_g \rangle = \omega_g(A)$. $\square$

Remark 3.4. Lemma 3.2 replaces the space-translation averaging used in [4, eq. (2.5)]: the set of limit points of $(\omega_g)_{g \in \mathcal{G}}$ is stable under translations by Lemmas 1.4 and 3.2, so *uniqueness of the limit already forces translation invariance*. This matters beyond aesthetics: averaging is precisely what produces a mixed limit in a multiple-phase regime, whereas Lemma 3.3 shows the unaveraged net is $\mathbb{Z}_2$ -invariant by itself.

4 The exact spectral identity

Proposition 4.1. *Let O be bounded open, $A, B \in \mathfrak{A}(O)$, $g \in \mathcal{G}$, and set $d := d(g, O)$, $M := \|A\| \|B\|$. Define the complex measure*

$$\rho_g(\cdot) := \langle A^* \Omega_g, P_{H(g)}(\cdot) B \Omega_g \rangle$$

on $\mathbb{R}$, and $G_g(z) := \int e^{izE} d\rho_g(E)$ for $\text{Im } z \geq 0$. Then

- (i) $\text{supp } \rho_g \subseteq [0, \infty)$ and $|\rho_g|(\mathbb{R}) \leq M$;
- (ii) G_g is holomorphic on $\{\text{Im } z > 0\}$, continuous and bounded by M on $\{\text{Im } z \geq 0\}$;
- (iii) $\omega_g(A \alpha_t(B)) = G_g(t)$ for every $|t| < d$;
- (iv) $\int_{\mathbb{R}} f(t) G_g(t) dt = 0$ for every $f \in L^1(\mathbb{R})$ with $\check{f}|_{[0, \infty)} = 0$.

Proof. (i) $\text{supp } \rho_g \subseteq \text{spec } H(g) \subseteq [0, \infty)$ by Fact 1.1. For a complex spectral measure, $|\rho_g|(\mathbb{R}) \leq \|A^* \Omega_g\| \|B \Omega_g\| \leq M$.

(ii) For $\text{Im } z \geq 0$ and $E \geq 0$ we have $|e^{izE}| = e^{-E \text{Im } z} \leq 1$; holomorphy and continuity follow from dominated convergence together with (i).

(iii) Let $|t| < d$. Then $g \equiv 1$ on $O_{|t|}$, so Fact 1.3(ii) applies to B and

$$\omega_g(A \alpha_t(B)) = \langle \Omega_g, A e^{itH(g)} B e^{-itH(g)} \Omega_g \rangle = \langle \Omega_g, A e^{itH(g)} B \Omega_g \rangle = \langle A^* \Omega_g, e^{itH(g)} B \Omega_g \rangle = G_g(t),$$

where the second equality uses $e^{-itH(g)} \Omega_g = \Omega_g$ and the last is the spectral theorem.

(iv) By (i) and $f \in L^1$, Fubini applies: $\int f(t) G_g(t) dt = \int_{[0, \infty)} \check{f}(E) d\rho_g(E) = 0$, since $\check{f}$ vanishes on $[0, \infty)$. $\square$

The cancellation $e^{-itH(g)} \Omega_g = \Omega_g$ in (iii) is the whole point: it is exact, and it is available because Ω_g is the ground state of the *same* operator that implements α_t locally. No approximation, and no error term, enters.

Theorem 4.2. *Let $(g_n)_{n \in \mathbb{N}} \subseteq \mathcal{G}$ be a sequence such that $d(g_n, O) \rightarrow \infty$ for every bounded open O , and suppose that for every bounded open O the restrictions $\omega_{g_n}|_{\mathfrak{A}(O)}$ converge in norm in $\mathfrak{A}(O)^*$. Let ω be the resulting state on $\mathfrak{A}$. Then ω is a ground state of $(\mathfrak{A}, \alpha)$ in the sense of Definition 2.1.*

Proof. (a) Fix $A \in \mathfrak{A}(O)$ and $t \in \mathbb{R}$. For n large, $d(g_n, O) > |t|$, so Lemma 3.1 gives $\omega_{g_n}(\alpha_t(A)) = \omega_{g_n}(A)$. Both $\alpha_t(A) \in \mathfrak{A}(O_{|t|})$ and $A \in \mathfrak{A}(O_{|t|})$, so letting $n \rightarrow \infty$ yields $\omega(\alpha_t(A)) = \omega(A)$. As $\bigcup_O \mathfrak{A}(O)$ is norm-dense in $\mathfrak{A}$ and $\|\alpha_t\| = 1$, (a) follows.

(b) Fix $A, B \in \mathfrak{A}(O)$ and put $M := \|A\| \|B\|$.

Continuity. Let $T > 0$ and let n be large enough that $d(g_n, O) > T$. For $|t| \leq T$ we have $A \in \mathfrak{A}(O) \subseteq \mathfrak{A}(O_T)$ and, by Fact 1.3(i), $\alpha_t(B) \in \mathfrak{A}(O_{|t|}) \subseteq \mathfrak{A}(O_T)$; hence $A \alpha_t(B) \in \mathfrak{A}(O_T)$ with $\|A \alpha_t(B)\| \leq M$, and therefore

$$\sup_{|t| \leq T} |\omega(A \alpha_t(B)) - \omega_{g_n}(A \alpha_t(B))| \leq M \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| \xrightarrow{n \rightarrow \infty} 0. \quad (1)$$

By Proposition 4.1(ii)–(iii), $t \mapsto \omega_{g_n}(A\alpha_t(B)) = G_{g_n}(t)$ is continuous on $[-T, T]$. Thus $t \mapsto \omega(A\alpha_t(B))$ is a uniform limit of continuous functions on $[-T, T]$, hence continuous there; as T was arbitrary it is continuous on $\mathbb{R}$, and bounded by M . In particular the integral below is well defined.

Vanishing. Let $f \in L^1(\mathbb{R})$ with $\check{f}|_{[0, \infty)} = 0$. For $T > 0$ set $\varepsilon(T) := \int_{|t| > T} |f(t)| dt$, so $\varepsilon(T) \rightarrow 0$ as $T \rightarrow \infty$. Fix $T > 0$ and decompose

$$\int_{\mathbb{R}} f \omega(A\alpha_t(B)) = \underbrace{\int_{|t| \leq T} f [\omega - \omega_{g_n}](A\alpha_t(B)) dt}_{(I)} + \underbrace{\int_{|t| \leq T} f \omega_{g_n}(A\alpha_t(B)) dt}_{(II)} + \underbrace{\int_{|t| > T} f \omega(A\alpha_t(B)) dt}_{(III)}.$$

By (1), $|(I)| \leq \|f\|_{L^1} M \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\|$, and $|(III)| \leq M\varepsilon(T)$ since $|\omega(A\alpha_t(B))| \leq M$.

For (II): choose n so large that $d(g_n, O) > T$. Then Proposition 4.1(iii) applies for all $|t| \leq T$, so, using Proposition 4.1(iv) and then (ii),

$$(II) = \int_{|t| \leq T} f G_{g_n} = \underbrace{\int_{\mathbb{R}} f G_{g_n}}_{=0} - \int_{|t| > T} f G_{g_n}, \quad |(II)| \leq M\varepsilon(T).$$

Letting $n \rightarrow \infty$ with T fixed kills (I) by hypothesis, and we obtain

$$\left| \int_{\mathbb{R}} f(t) \omega(A\alpha_t(B)) dt \right| \leq 2M\varepsilon(T) \quad \text{for every } T > 0.$$

Letting $T \rightarrow \infty$ gives 0. Finally, for general $A, B \in \mathfrak{A}$ choose local $A_k \rightarrow A$, $B_k \rightarrow B$ in norm; the map $(A, B) \mapsto \int f \omega(A\alpha_t(B))$ is jointly norm-continuous with bound $\|f\|_{L^1} \|A\| \|B\|$, so (b) extends to $\mathfrak{A}$. $\square$

Corollary 4.3. *Assume the local number bound*

$$\sup_{g \in \mathcal{G}} \omega_g(N_{\tau, B}) \leq c(|B|) < \infty \quad (\text{H2})$$

so that [4, Thm. 2.3] applies. Then sequences as in Theorem 4.2 exist: any sequence $(g_n) \subseteq \mathcal{G}$ with $d(g_n, O) \rightarrow \infty$ for all O admits a subsequence along which $\omega_{g_n}|_{\mathfrak{A}(O)}$ converges in norm for every bounded open O , and every resulting limit is a locally normal ground state of $(\mathfrak{A}, \alpha)$.

Proof. Fix an exhausting sequence $O_1 \subseteq O_2 \subseteq \dots$ of bounded open sets with $\bigcup_k O_k = \mathbb{R}$. By [4, Thm. 2.3] and (H2), $\{\omega_{g_n}|_{\mathfrak{A}(O_k)}\}_n$ is contained in a norm-compact subset of $\mathfrak{A}(O_k)^*$ for each k ; norm-compact metric spaces are sequentially compact, so a diagonal argument produces a subsequence converging in norm on every $\mathfrak{A}(O_k)$, hence on every $\mathfrak{A}(O)$ by isotony. Limits of norm-convergent sequences of normal states are normal [4, Thm. 2.3]. Theorem 4.2 applies. $\square$

Corollary 4.4 (reduction of the target). *Assume (H2). If $(\mathfrak{A}, \alpha)$ has a unique locally normal ground state ω_∞ which is $\mathbb{Z}_2$ - and translation-invariant, then*

$$\omega_g \longrightarrow \omega_\infty \quad \text{in norm on every } \mathfrak{A}(O),$$

along the full net $\mathcal{G}$.

Proof. Suppose not; then there are O , $\varepsilon > 0$ and a sequence g_n , cofinal in $\mathcal{G}$, with $\|(\omega_{g_n} - \omega_\infty)|_{\mathfrak{A}(O)}\| \geq \varepsilon$. By Corollary 4.3 pass to a subsequence converging in norm on all local algebras to a locally normal ground state ω_2 ; by Lemmas 3.2, 3.3 and Remark 3.4, ω_2 is $\mathbb{Z}_2$ -invariant, and by hypothesis $\omega_2 = \omega_\infty$. This contradicts $\|(\omega_2 - \omega_\infty)|_{\mathfrak{A}(O)}\| \geq \varepsilon$. $\square$

5 Remarks

Remark 5.1 (why the generator form is unusable). The characterisation of ground states by $-i\omega(A^*\delta(A)) \geq 0$ requires $t \mapsto \alpha_t(A)$ to be norm-differentiable on a dense *-subalgebra. For a Weyl algebra this fails badly: $\|W(f, h) - W(f', h')\| = 2$ whenever $(f, h) \neq (f', h')$, so α is not even norm-continuous and no densely defined norm-generator exists. Definition 2.1 and Proposition 4.1 avoid generators entirely, working only with bounded operators and the unitary group $e^{itH(g)}$. As a byproduct, the domain hypothesis “ $A\Omega_g \in D(H(g)^{1/2})$ ”, which a generator-based argument would need, is not required.

Remark 5.2 (M0 needs P2: the two pillars are not independent). Theorem 4.2 assumes *norm* convergence on local algebras, not merely weak-* convergence of (ω_{g_n}) . This is not a technical convenience. In the proof, term (I) must be shown to vanish, and weak-* convergence gives only pointwise convergence in t of the integrand, with no dominated convergence theorem available for nets. Norm convergence on $\mathfrak{A}(O_T)$ supplies exactly the uniformity in $t \in [-T, T]$ that is needed. Hence the compactness of [4, Thm. 2.3] enters the argument *twice*: once to produce a limit, once to pass the spectral condition to it.

Remark 5.3 (what is *not* proved here). Theorem 4.2 does not assert that a limit exists along the full net; it identifies the nature of every limit point. Nor does it produce a rate. The remaining inputs are: (H2) without space-averaging (Milestone M1); uniqueness of the locally normal $\mathbb{Z}_2$ - and translation-invariant ground state of $(\mathfrak{A}, \alpha)$ (Milestone M3, or the uniform-gap substitute); and, for rates, a uniform spectral gap together with a quasi-adiabatic argument exploiting the exact light cone of Fact 1.3(i) (Milestone M4).

Remark 5.4 (what M0 buys). Rosen’s proof that $\omega_g \rightarrow \omega_\infty$ in the quadratic model [9, §5] has two steps: identify the limit, then upgrade the topology using [4, Thm. 4.1]. In the quadratic model the first step is free because $\mu_g \rightarrow \mu_1$ in the one-particle space. For $P(\varphi)_2$ it is not, and this is exactly what Guerra–Rosen–Simon list as open [7, Problems 1–2]. Theorem 4.2 supplies that first step in a form that is exact and independent of any Euclidean input, and thereby reduces the removal of the spatial cutoff to a *uniqueness* question, where the post-1989 literature has delivered.

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